

DAY 4 HOMEWORK

Plan Your First AI Application Before Writing Python

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Date: 8/6/26

Project/App Name: MediMind

Assignment: Complete all eight planning items for your own AI capstone idea. Do not begin coding until your plan is specific, testable, and easy to explain.

Submission: Submit this completed document and upload it to your GitHub project folder using a clear filename such as day4_ai_app_plan.docx.

1. Clear Problem Statement

Write one specific sentence using this structure: "Many [users] struggle with [specific problem] because [reason]."

Helpful reminder: Describe the problem, not the features of your app. Avoid broad words such as "people," "things," or "everything."

My problem statement:

Many patients and caregivers struggle with remembering to take their medication because it is easy to forget to take it.

2. User Story

Complete: "As a [type of user], I want to [action], so that [benefit]."

My user story:

As a someone who takes or administers medication, I want to be able to remember when to take it so that my health remains well.

3. Input–Process–Output (IPO) Chart

Input is what the user gives the app. **Process** is what the app does. **Output** is what the user receives.

INPUT	PROCESS	OUTPUT
Medication Information	Creates a time bracket of when the medication has to be administered. Sets an alarm for the time that it should optimally be administered.	An alarm that rings and verifies you take your medication.
Calendar/Schedule	Finds a time that the medication should be administered in the bracket that also fits with their schedule.	Sets an alarm for the best time.

4. Pseudocode

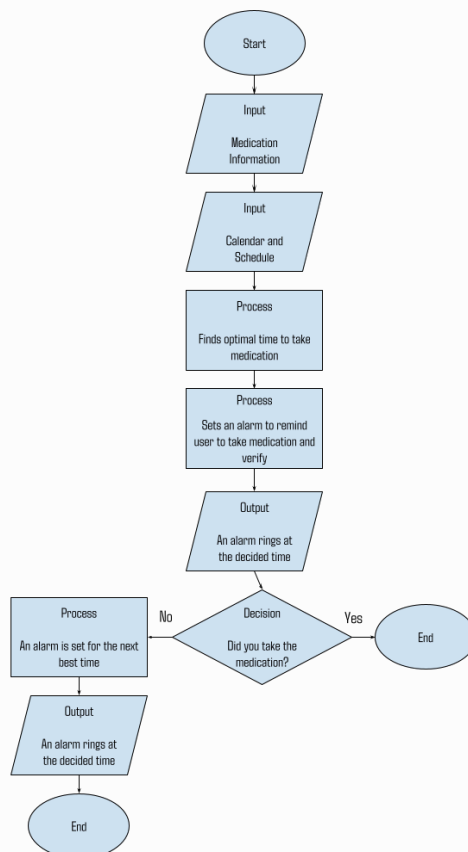
Write your program plan in plain English. Include: START, user input, validation, processing, output, and END. Use IF/ELSE when the app must make a decision.

1	START
2	User inputs their medication information
3	User inputs their calendar
4	AI creates a time bracket for when the medication should be administered
5	AI checks for a time that fits users schedule and fits in the time bracket
6	An alarm is set for the best time
7	The alarm rings and verifies the user takes the medication
8	If user cannot take medication at the time the AI will set an alarm for the next best time
9	END
10	else END

5. Flowchart

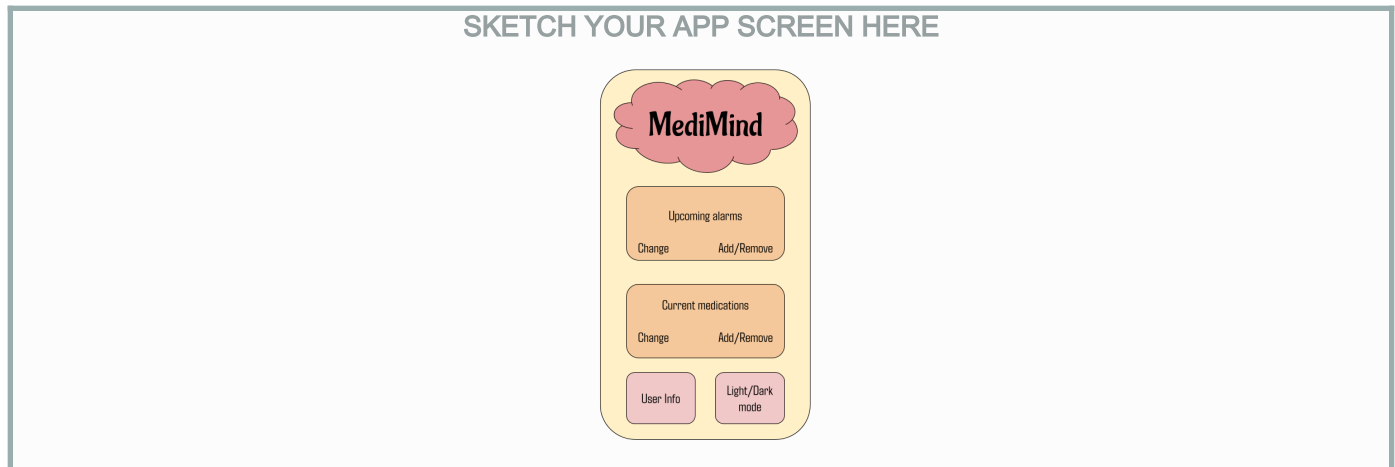
Draw the order of your app. Include at least: Start, Input, one Decision, Process, Output, and End. Label each arrow clearly.

DRAW YOUR FLOWCHART HERE



6. User Interface Sketch

Sketch what the user will see. Include: app title, input areas, button(s), warning/error location, and output area.



7. Two Test Cases

For each test, show exactly what the user enters and what the app should return. Use one normal test and one edge case, such as empty input, unclear wording, or missing information.

Test Case 1

Test purpose	Checking if they can set an alarm with an event on the calendar.
Exact input	Medication: Insulin Pill When to take: Take with dinner Calendar: Dinner with friend
Expected output	An alarm before the event starts reminding them to take their medication with their dinner and verifying they took when the event ends.
How I will know it passed	The alarm rings before the event starts - not during. Doesn't send any notifications until the event is over.

Test Case 2

Test purpose	Checking if they can set and alarm for a medication without a clear time given
Exact input	Medication: Insulin pill When to take: Once a day Calendar: Several meetings
Expected output	They ask for a more specific time and if not given they set it at the beginning or end of the day during no events.
How I will know it passed	The alarm doesn't ring at a wrong time.

8. Scope Statement: Version 1

Keep your first version small. List only the most important features under "Will Do." Put future ideas and complicated integrations under "Will Not Do Yet."

VERSION 1 WILL DO	VERSION 1 WILL NOT DO YET
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Remind when to take medication	Verifying medication information
Check with calendar for availability	Tell when medication needs to be refilled
Verify user takes medication	

One-Breath Project Explanation

Explain your project in 2–3 sentences: who it helps, what information it uses, what it does, and what result it provides.

MediMind helps patients and caregivers remember when to take or administer medication. It sets alarms by accessing your medical information and calendar then verifies you take the medication.

Submission Checklist

- ☐ I completed all eight sections.
- ☐ My problem statement is specific.
- ☐ My input, process, and output match each other.
- ☐ My pseudocode and flowchart show the same sequence.
- ☐ My interface sketch includes input, button, messages, and output.
- ☐ My two test cases include exact input and expected output.
- ☐ My Version 1 scope is small enough to build.
- ☐ I reviewed spelling and clarity.
- ☐ I saved the file with a clear lowercase filename.
- ☐ I uploaded the completed document to GitHub.

Teacher Review (40 points)

Required Item	Points	Teacher Notes
1. Problem statement	/ 5	
2. User story	/ 5	
3. IPO chart	/ 5	
4. Pseudocode	/ 5	
5. Flowchart	/ 5	
6. Interface sketch	/ 5	
7. Two test cases	/ 5	
8. Scope statement	/ 5	